

Do not cry over repositioning.

I never assume my first placement is perfect. Even if everything looks good initially, the layout may not be optimised. I may need to rotate components, shift parts, or completely reposition sections. This is normal. Repositioning is part of the process. I do not resist it. I plan for it.

Look out for collisions and

clearances. If design rules are not set correctly, components can overlap at the footprint level, making soldering impossible. Connectors can collide with the enclosure. Components can violate edge or height constraints. These issues are avoidable if I continuously check for collisions and clearances throughout the layout process.

Read and follow layout guidelines. Layout guidelines are not optional. We once had an analogue IC in an MPPT solar charge controller design. The schematic was correct. The PCB did not work. Nothing in the circuit had changed. In the next revision, we followed the IC layout guidelines exactly. No schematic changes. The board worked perfectly. That experience reinforced one thing for me. Reading and following layout guidelines is critical, especially for analogue, power, and high-frequency designs.

Documentation: The key to a perfect build

I make sure to provide clear and complete documentation to everyone involved in the project. This includes the PCB fabricator, the assembly house, and all internal

teams. Documentation is not meant only for manufacturing. It is equally critical for coordination within the organisation.

The software team needs clear connector pinouts and well-defined hardware-software interface diagrams. Any interface that touches firmware must be documented properly. These documents should be clean, unambiguous, and shared with all teams through proper channels. Good documentation removes assumptions, reduces confusion, and prevents rework later in the project.

After architecture, component selection, schematics, layout, and documentation, there is one mindset that matters more than any technical step.

Not to believe in yourself. This may sound contrary to common motivational advice, but from a PCB design perspective, this is something we strongly believe in. I may feel confident that my design is perfect, but in reality, I am the most likely person to miss my own mistakes. Familiarity creates blind spots. That is why the next step is critical.

Get judges for your work. Not random people, but technically strong engineers who can objectively review what I have done. Engineers who understand hardware, layout, manufacturing, and system-level behaviour. Fresh eyes catch problems that I will never see.

Review, review, and review your design. Reviews should not happen only at the end. They must start from the beginning. Review the architecture. Review the symbols.

Review the footprints. Review the schematics. Review the layout. Review the documentation. Every stage needs scrutiny. This is how robust and reliable designs are built.

Define and follow at all costs

We are going to define many rules, procedures, and processes to achieve good design practices. But defining them and keeping them aside does not help since defining without following is meaningless. If I define something, I must follow it. And if there is leadership in the room, leaders must allow their teams to follow these processes as well. When leaders define rules but do not follow them themselves, the team gets demotivated.

If a process feels slow or inefficient, the right approach is to sit with the team, review it, and improve it. Make it more efficient if required. But do not define processes and then prevent people from following them.

As engineers, once we define something, we must follow it at all costs. Even if it feels like we are saving time by skipping a step, we are not. That time will come back later, when the board is in our hands, and we are debugging problems. So define, and follow at all costs.

With that, let us go build some solid systems. **EFY**

The article is based on a talk called 'Hardware Design Considerations for First Time Right Designs' at EFY Expo in India Electronics Week, featuring the opening speech by Chetan Soni, Senior Engineer, NXP Semiconductors. It has been transcribed and curated by Saba Aafreen, Electronics For You.

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